

Multiple Choice (10 marks)

1. This biome has cold winters and is known for its pine forests.
 - (a) Tundra
 - (b) Rainforest
 - (c) Grassland
 - (d) Chaparral
 - (e) Savanna
2. Which of these is the smallest unit that natural selection can change?
 - (a) an individual's phenotype
 - (b) an individual's genotype
 - (c) a population's phenotype
 - (d) a species' genome
 - (e) a species' genotype
3. Why does milk "when kept in a refrigerator?
 - (a) Lactose crystallization at low temperatures causes milk to spoil
 - (b) Milk spoils due to absorption of odors in the fridge
 - (c) Milk spoils due to the presence of mesophiles
 - (d) Milk spoils due to chemical reactions in the fridge
 - (e) Thermophiles cause milk to spoil in the refrigerator
4. Over the past two decades retroviruses have profoundly im-pacted the growing field of molecular biology. What retroviral enzyme has most revolutionized this field?
 - (a) RNase H
 - (b) Helicase
 - (c) integrase
 - (d) DNA ligase
 - (e) protease
5. Chemical substances released by organisms that elicit a physiological or behavioral response in other members of the same species are known as
 - (a) auxins
 - (b) neurotransmitters

- (c) enzymes
- (d) hormones
- (e) pheromones

True / False (10 marks)

Answer True or False

6. True or False: Amphibians were the first group of animals to evolve lung tissue that could fully support their respiratory requirements independently.
7. True or False: The probability that the fourth child will be male is $1/2$, as gender is determined independently of previous children.
8. True or False: A photosynthetic plant in iron- and magnesium-deficient soil will exhibit dark, large, and robust growth despite nutrient limitations.
9. True or False: Acetylcholine does not function as an intracellular messenger in cellular signaling pathways.
10. True or False: Approximately 50% of the human genome is estimated to encode proteins.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Discuss the concept of heritability in the context of a population of *Drosophila melanogaster*, where the mean number of abdominal bristles shifts from 38 to 40.6 after selective breeding. Calculate the heritability of this trait and explain its significance in understanding genetic inheritance.
12. Discuss the significance of stem reptiles in the context of evolutionary biology, focusing on their unique evolutionary lineage leading to modern reptiles.

End of Paper